

# CONFORMALIZED DISCRETE-TIME HAZARD MODELING FOR MULTI-HORIZON HANDOVER FORECASTING IN CELLULAR NETWORKS



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## ABSTRACT

- **Problem:** LTE handovers operate reactively via Event A3, initiating target cell preparation only after link degradation has already occurred. This causes radio-link failure (RLF) during rapid channel drops and induces high ping-pong rates.
- **Logging Artifact Discovery:** Physical alignment audit reveals that standard 1 Hz diagnostic logs write post-handover cell state before the RRC command in 76% of events. Failing to lag features produces a spurious +235% AUPRC inflation (0.179 -> 0.600) prevalent in un-audited literature.
- **Hazard Formulation:** Framing handover forecasting as discrete-time survival analysis with a single gradient-boosted tree model guarantees coherent multi-horizon ordering without post-hoc heuristics, reaching AUPRC 0.179 at 1 s (2.6x over baseline) and 0.475 at 5 s.
- **Risk Guarantees:** Conformal Risk Control certifies an early-warning alarm threshold that strictly bounds per-drive miss rate  $\leq 0.20$  (realised 0.168 at 47% alarm rate). Over handovers, 36.4% of events are fundamentally bounded by the 2 s post-handover blanking policy limit.

## BACKGROUND

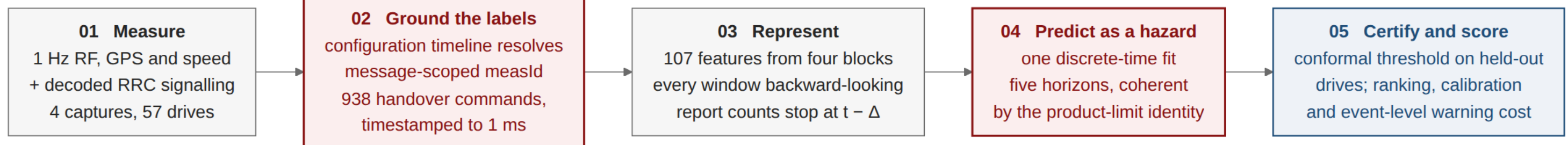
- **3GPP LTE Mobility Mechanism:** Handover decisions are governed by RRC MeasurementReport messages. Event A3 fires when neighbor RSRP exceeds serving by an offset ( $M_n - Of_n > M_s + Of_s + Off$ ) for a continuous Time-To-Trigger (TTT), creating an unavoidable delay.
- **Signalling Inefficiencies:** In dense urban topologies, 26.3% to 43.8% of handovers return to the source cell within 15 seconds, expending unnecessary signaling load, radio resource control overhead, and UE battery consumption.
- **Literature Evaluation Pitfalls:** Audit of 22 published models indicates widespread data leakage: random row-level splitting across continuous drive tests inflates AUPRC by up to 48% and misranks models relative to cluster-safe cross-campaign splits.

## OBJECTIVES

- **O1 - Alignment Audit:** Identify and correct physical logging/export latency between handset traces and decoded RRC event clock.
- **O2 - Leakage-Free Benchmarking:** Establish a rigorous Leave-One-Campaign-Out (LOCO) protocol that prevents temporal and spatial information leakage.
- **O3 - Monotone Multi-Horizon Forecasting:** Develop a discrete-time survival/hazard framework guaranteeing mathematically ordered cumulative incidence (0.5 s to 5 s).
- **O4 - Distribution-Free Risk Control:** Formulate Conformal Risk Control (CRC) on clustered drive units to certify operational alarm thresholds with statistical bounds.
- **O5 - Operational Characterization:** Decode live RRC signaling timelines to quantify Event A3 conversion efficiency and ping-pong sensitivity under live deployments.

## PROPOSED METHODOLOGY

- **Signalling Decoding:** Replay RRC connection reconfiguration timelines under TS 36.331; ground truth extracted from mobilityControlInfo with 100% confirmed completion.
- **Causal Feature Engineering:** 112 strictly causal features: 89 Radio (serving RF + projected neighbor RSRP via zero-order hold), 17 Mobility (GPS speed, acceleration, bearing), 6 History (dwell time, recent handover count). All export features lagged 1 row.
- **Hazard Formulation:** Single LightGBM model estimates interval hazard  $\lambda_k(x_t)$ . Cumulative incidence derived via product-limit identity  $F_k(x_t) = 1 - \prod_{j=1}^k (1 - \lambda_j(x_t))$ , guaranteeing monotonic ordering without heuristics.
- **Conformal Risk Control:** Certify threshold  $\lambda^*$  on calibration sessions such that expected per-unit loss satisfies  $E[L] \leq \alpha$  with distribution-

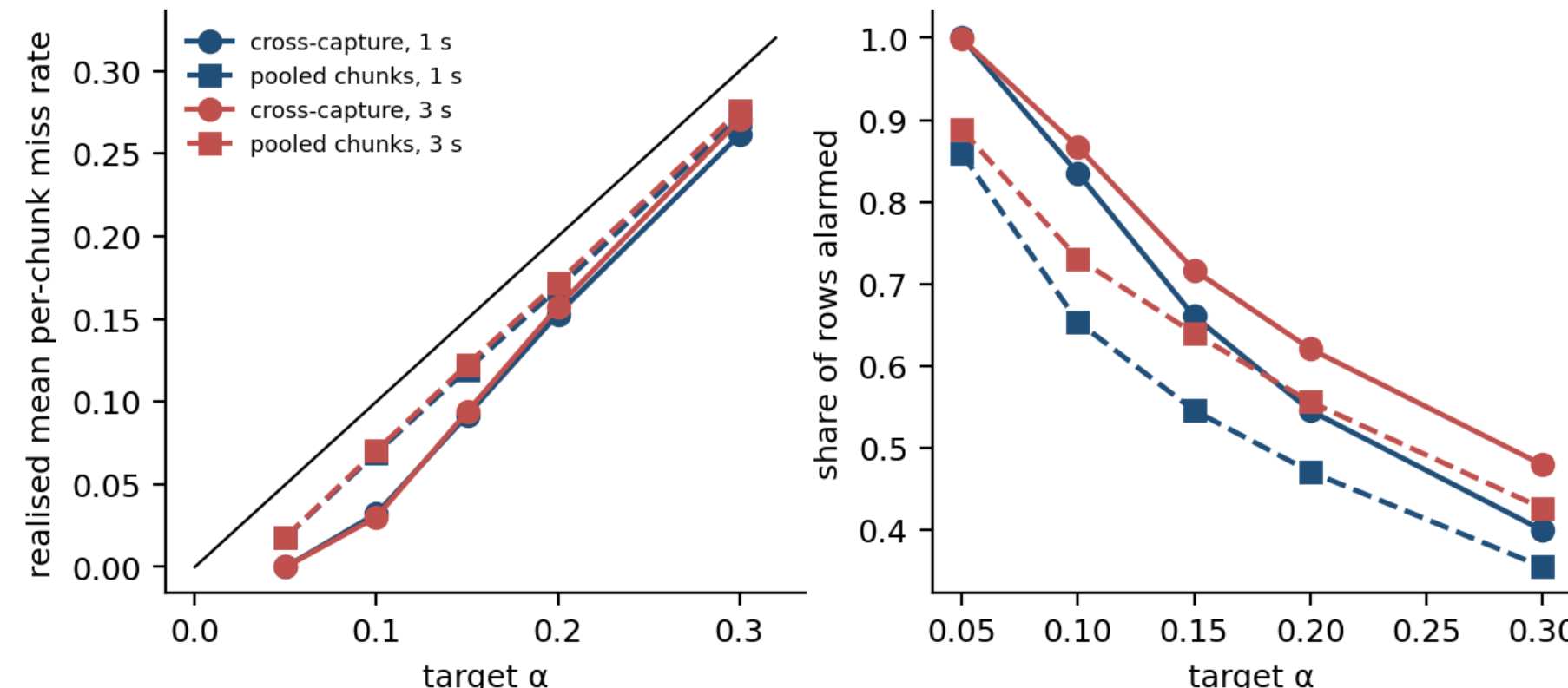
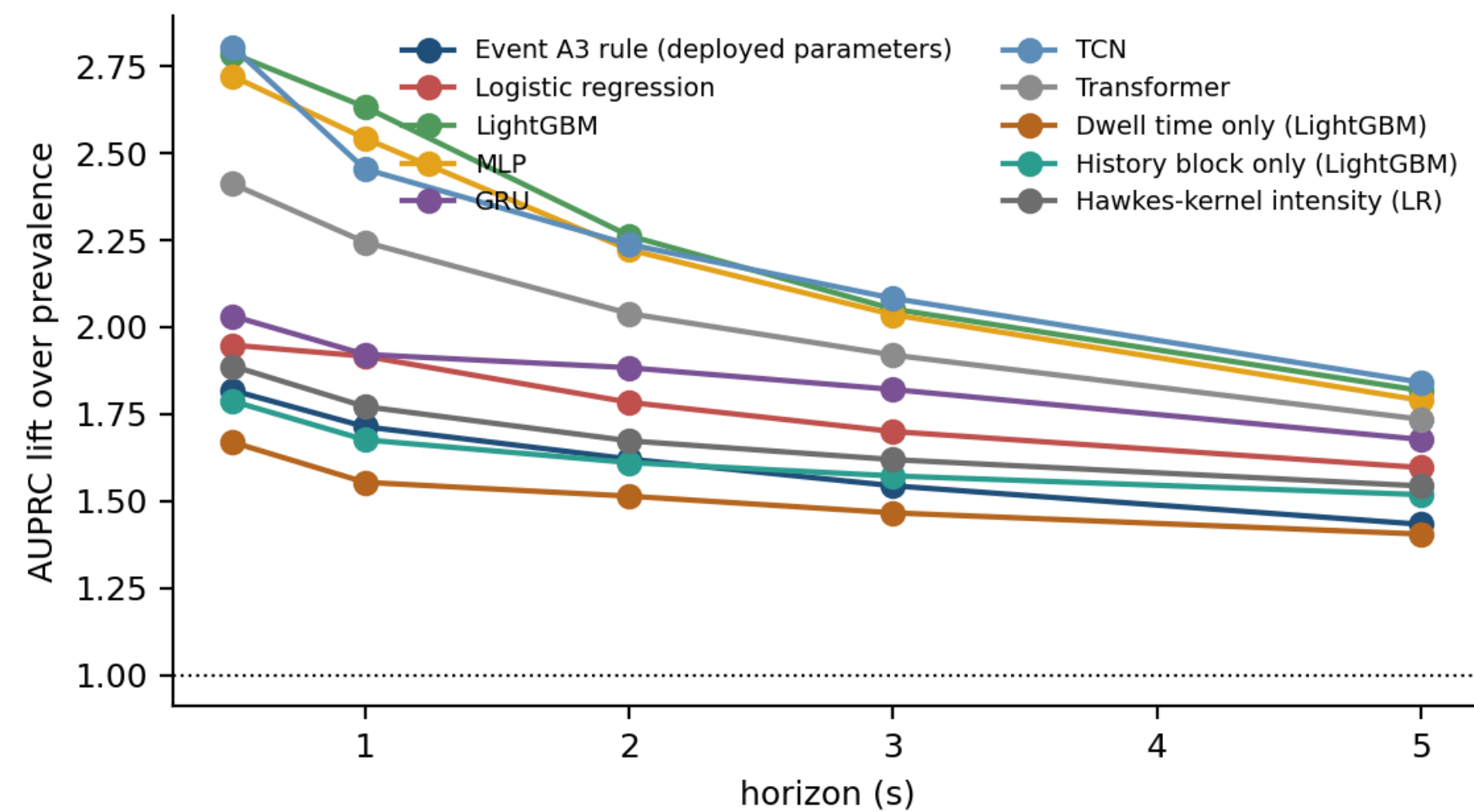


## EXPERIMENTAL DESIGN

- **Measurement Testbed:** Commercial live LTE network across Dhaka and Gazipur, Bangladesh, captured using an instrumented vehicle with XCAL diagnostic drive-test hardware.
- **Four Continuous Campaigns:** Campaign 1 (Urban Arterial, 45 min, 39.1 km/h), Campaign 2 (Urban Loop, 24 min, 25.0 km/h), Campaign 3 (Dense Urban, 60 min, 21.4 km/h), Campaign 4 (Highway, 42 min, 50.9 km/h).
- **Corpus Scale:** 96 km total distance, 10,260 1-Hz measurement rows (8,586 usable modelling rows after 2-second blanking & quality control), 957 confirmed handover commands, and 341 RRC re-establishments.

## KEY OUTCOMES/RESULTS

- **Predictive Superiority:** Discrete-time LightGBM hazard achieves AUPRC 0.179 at 1 s (2.6x over 6.8% base rate), 0.252 at 2 s, and 0.475 at 5 s, outperforming deployed Event A3 baseline (0.116 at 1 s) and deep sequence models (GRU: 0.158, Transformer: 0.155).
- **100% Horizon Monotonicity:** Hazard formulation yields exactly 0.0% horizon ordering violations, whereas independent binary classifiers produce 40.2% contradictory violations across horizons.
- **Conformal Risk Verification:** At target  $\alpha = 0.20$ , certified warning threshold achieves realised mean per-unit loss of 0.168 (47% alarm rate) under exchangeable pooling and holds in 83% of cross-campaign holdout rotations.
- **Network Operating Realities:** 74.8% of A3 handovers fire under positive offset (+1 dB dominant). Between 63% and 73% of triggered A3 episodes are declined by the eNodeB without a handover command.



## NOVELTY/SIGNIFICANT CONTRIBUTIONS

- **1. Diagnostic Alignment Audit:** First systematic demonstration that commercial diagnostic 1 Hz exports suffer a write-before-read logging artifact, removing spurious results and establishing a reproducibility standard.
- **2. Coherent Hazard Framework:** First unified discrete-time survival formulation applied to multi-horizon mobility prediction, eliminating horizon incoherence by construction.
- **3. Clustered Conformal Risk Guarantees:** First deployment of Conformal Risk Control on drive-test sessions, testing exchangeability across routes and revealing the 36.4% evaluation policy limit under post-handover blanking.
- **4. 22-Model Meta-Audit:** Screened 108 papers and audited 22 models, demonstrating that random-row evaluation inflates scores by up to 48% and corrupts comparative leaderboards.

## COMMERCIAL VIABILITY

- **Zero Hardware Changes:** Purely software-based inference executable on handset baseband chipsets or integrated as an auxiliary background telemetry daemon.
- **Open-RAN (O-RAN) xApp Compatibility:** Easily packaged as a near-Real-Time RIC xApp or non-RT rApp, receiving standard E2 measurement reports to guide proactive Traffic Steering (TS) and QoS optimization.
- **Signalling Overhead & Energy Reduction:** Early target cell preparation mitigates user-plane interruption and eliminates up to 26.3% wasteful ping-pong executions, saving base-station capacity and UE power.

## OBE ATTRIBUTES

| POs  | Tick | Ks | Tick | Ps | Tick | As | Tick |
|------|------|----|------|----|------|----|------|
| PO1  | ✓    | K3 | ✓    | P1 | ✓    | A1 | ✓    |
| PO2  | ✓    | K4 | ✓    | P2 | ✓    | A2 | ✓    |
| PO3  | ✓    | K5 | ✓    | P3 | ✓    | A3 | ✓    |
| PO4  | ✓    | K6 | ✓    | P4 | ✓    | A4 | ✓    |
| PO5  | ✓    | K7 | ✓    | P5 | ✓    | A5 | ✓    |
| PO6  | ✓    | K8 | ✓    | P6 | ✓    |    |      |
| PO7  | ✓    |    |      | P7 | ✓    |    |      |
| PO8  | ✓    |    |      |    |      |    |      |
| PO9  |      |    |      |    |      |    |      |
| PO10 | ✓    |    |      |    |      |    |      |
| PO11 | ✓    |    |      |    |      |    |      |
| PO12 |      |    |      |    |      |    |      |

- **PO3 / PO4 / PO5:** Design of experiments on live operator LTE network; rigorous multi-campaign data analysis; modern machine learning and diagnostic toolchain.
- **PO6 / PO7 / PO8:** Operator anonymity preserved; societal impact through signaling energy waste characterization; professional ethics in reporting negative results.
- **PO10 / PO11:** Exhaustive documentation across 7 chapters; engineering management and resource allocation over two academic terms.
- **K3 / K4 / K5 / K6:** Mathematical survival analysis fundamentals; specialized RRC decoding under 3GPP TS 36.331; causal feature design; live-network instrumentation.
- **K7 / K8:** Societal role of mobile broadband; critical synthesis of 108 papers and 22 audited models.
- **Complex Problems (P1-P7, A1-A5):** Resolution of conflicting alarm-vs-miss trade-offs; non-standard evaluation procedures; integration across radio propagation, survival theory, and cellular control plane.

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### OPEN-SOURCE BENCHMARK & REPRODUCIBILITY ARTIFACTS

- **Causal Signalling & Telemetry Pipeline:** Complete Python framework, millisecond-precision 3GPP TS 36.331 RRC protocol decoders, 112 strictly causal radio and mobility feature builders, and automated verification testbenches.
- **Conformal Calibration & Model Checkpoints:** Production LightGBM hazard models, Leave-One-Campaign-Out cross-validation harnesses, and distribution-free CRC calibration algorithms open-sourced under Apache-2.0.

